

Lab #4 – Autonomous Robot Motion Control with Sensors

Thomas Parsley & Nick Platon

1. Exercise 1

1.1 Circuit:



Figure 1.1.1 - Raspberry Pi Connected with Kobuki Robot

1.1 Summary:

In Exercise 1, we learned how to read data from the built in sensors on the Kobuki Robot. The Kobuki robot serially sends a basic sensor data packet through its data port. Using the RaspberryPi, we can read the values of this data packet. Using this we can get information about the bumper states, cliff sensor states, and wheel drop states. This exercise builds a foundation for creating mobile robots, and will be used in the following lab exercises.

2. Exercise 2.1

2.1 Circuit:

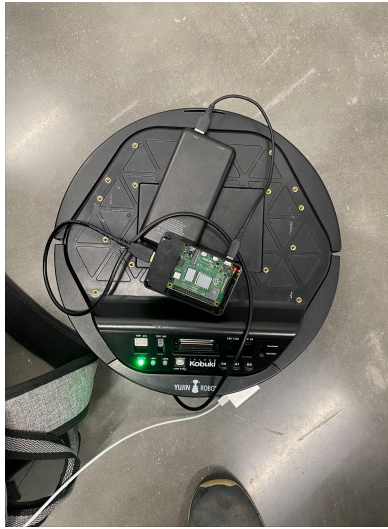


Figure 2.1.1 - Raspberry Pi Connected with Kobuki Robot

2.1 Summary:

In Exercise 2, we used the sound sensor to navigate the Kobuki Robot through an obstacle course. Using the information learned in previous labs, we measured the distance from the sound sensor, located at the front of the robot, to the nearest object, usually being the wall of the obstacle course. Once we reached a close distance, the Kobuki robot would turn the correct way and repeat until the obstacle course was finished.

3. Exercise 3.1

3.1 Circuit:



Figure 3.1.1 - Raspberry Pi Connected with Kobuki Robot

3.1 Summary:

In Exercise 3, we used all the built in sensors on the Kobuki Robot to allow the robot to continuously move throughout the room, without getting stuck. Utilizing our knowledge on reading sensor inputs from Exercise 1, we read each sensor and changed the robots direction depending on the sensor activated. The program would back up the robot and turn in the direction opposite of the sensor activated. This allowed the robot to turn away from an obstacle and move to other parts of the room, or avoid falling off ledges.

4. Supplemental Questions

1. Briefly summarize what you learned from this lab?

In Lab 4, we learned how to utilize sensor data to affect the course of the Kobuki Robot. Initially, we learned to read the basic sensor data packets that are serially transmitted from the Kobuki Robot to the RaspberryPi. This transmitted data about the bumpers, cliff sensor, and wheel drop.

Then we used the sound sensor to gather distance data in order to navigate through a specially designed obstacle course. Using information learned from previous labs, we used the sound sensor to read the distance from the front of the Kobuki robot to the nearest wall of the obstacle course. Once close enough to the wall, we would turn the robot and repeat the cycle to complete the obstacle course.

Finally, we used all sensors to create a feedback loop that would allow the Kobuki robot to continuously move throughout the room. The robot would move forward until a sensor is activated, where it would back up and turn slightly. This would repeat until the object was no longer in the path of the robot.

This lab was important in understanding mobile robotics design. Mobile robots have to read sensor data to gather information about the world around them and act on that information to complete their purpose. This is exactly what we did in Exercise 2 and Exercise 3, where the Kobuki Robot had to respond to its environment to navigate through the obstacle course or around the room.

2. If you received out-of-range or missing values from the sonar sensor what may be the reason for this?

If the signal response takes longer than the maximum pulse width, the sonar sensor will not be able to read the time when the signal returns. This would cause the sound sensor to either read a value that is out of range or return a missing value. The maximum pulse width of the sensor is 18.5ms, meaning that the time it would take to reach the Kobuki Robot from the object would have to be less than that. In the opposite direction, if the signal response is too quick, the

sound sensor may not be in input mode yet, and will miss the return signal completely.

3. Summarize the capabilities of sonar, bumper, wheel drop, and cliff sensor in robot navigation.

The sonar sensor can be used to measure the distance between itself and the closest object in front of it. It does this by sending out a signal and measuring the time it takes to come back to the sonar sensor. The RaspberryPi sends the signal to the sound in output mode. Then after some time, it switches to input mode and reads the signal once it comes back. Using the speed of sound, the distance is calculated. This can be used to change direction whenever something is too close to the robot.

The bumper at the front of the robot shows it is activated whenever it has been pressed. There are three parts to the bumper, the left, middle, and right. Whenever one is activated, a specific value representing that part of a bumper is sent as the bumper data, 1, 2, and 4 respectively. If more than one is sent. The values are added together. This can be used to change the robot's direction whenever the Kobuki runs into an object.

The wheel drop sensor sends data whenever one of the wheels is dropped. If the left wheel is dropped, it sends a value of 1. If the right wheel is dropped, it sends a value of 2. If both are dropped it sends 3. This can be used to adjust the robot's path if the wheel has dropped.

The cliff sensors are sensors that send a data value whenever the section the sensor is in is over a cliff. There are three sensors, left, middle, and right, sending values 1, 2, or 4, respectively. This sensor can be used to make sure the robot does not fall off of any ledges, such as tables or stairs.

ACKNOWLEDGMENTS

I certify that this report is my/our own work, based on my/our personal study and/or research and that I/we have acknowledged all material and sources used in its preparation, whether they be books, articles, reports, lecture notes, and any other kind of document, electronic or person communication. I/We also certify that this assignment/report has not previously been submitted for assessment anywhere, except where permission has been granted from the coordinators involved.

Author-1 Signature Thomas Grealy

Author-2 Signature Alata